Fast and Slow Responses to AI Competition: How Internal Generative AI Reshapes Human Contribution in Online Q&A Communities

Abstract

The advent of generative artificial intelligence (GenAI) has disrupted online question-and-answer (Q&A) communities and prompted debates on managing AI and human contributions. Unlike prior studies on external general-purpose GenAI systems like ChatGPT, our work analyzes how an additional platform-provided, internal GenAI changes Q&A communities. Grounded in dual-process (fast-and-slow) theory and situated in a competitive setting, we argue that the total AI effect is not one deterrence effect but the sum of two separable components: a fast AI label effect and a slower AI content effect. Using a randomized field experiment at a leading technical Q&A community, we examine two outcomes. For answer supply, the AI label invites human contribution because a skeptical community treats the AI as a weak competitor, whereas longer AI content signals coverage and suppresses it, yielding a negative average effect. For answer quality, higher-quality AI raises the quality of human answers as contributors improve upon a stronger benchmark. Decomposing the total AI effect explains why aggregate effects diverge from either component and offers a theory-driven framework for predicting human responses as AI capabilities advance. We contribute to the emerging literature on GenAI's impact by differentiating AI's impact on different decision signals, which provides practical insight for platforms integrating GenAI into online Q&A communities.

Keywords: Generative AI, Human-AI Interactions, Online Communities, Dual-Process Theory, AI Label, AI Content.

1. INTRODUCTION

Generative artificial intelligence (GenAI) and large language models (LLMs) have transformed online information-seeking behavior and present a significant challenge to traditional online question-and-answer (Q&A) communities. Recent studies document a decline in user participation on these platforms after the introduction of ChatGPT (Quinn & Gutt 2025; del Rio-Chanona et al. 2024; Sanatizadeh et al. 2025; Xue et al. 2026), which raises concerns about the future viability of online Q&A communities.

In response to the emergence of GenAI, many online Q&A platforms, such as Quora,[1] have now developed their own LLMs that leverage their historical data repositories to answer questions. This integration of customized GenAI within communities (hereafter "internal GenAI;" see Figure 1) aims to enhance user experience through quick response and to recapture question demand in an environment where external GenAI tools remain widely accessible. This setting reflects the practical reality facing online Q&A platforms: because they cannot eliminate external GenAI usage, building an internal GenAI system becomes the one strategic action that is simultaneously forced upon them and fully within their power to design.

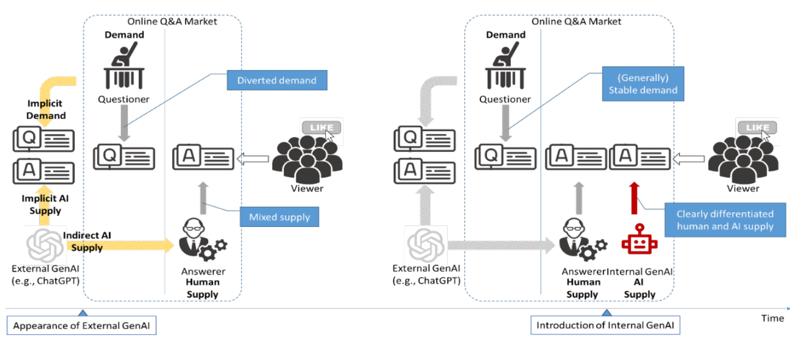

Figure 1. Difference between external GenAI and internal GenAI in online Q&A communities

[1] Further information about Quora's AI integration can be found at: https://quorablog.quora.com/Poe-1?ch=10&oid=98806369&share=e6baeb55&srid=C5jip&target_type=post.

Internal GenAI is not external GenAI relocated inside the platform. It is a distinct sociotechnical and governance arrangement, and this distinction is what makes the phenomenon theoretically important. As the left panel of Figure 1 shows, external GenAI tools operate independently of any platform. On the demand side they divert questions away from Q&A communities. On the supply side, human answerers can consult them privately when they write responses. This private use leaves no visible trace and blurs the line between human and AI contributions. Internal GenAI operates under a different set of rules. The platform controls it directly, labels its output explicitly as AI-generated, and places it in the first-answer position (Figure 1, right panel). These features change the legitimacy and accountability of the AI answer within the community. The AI answer is no longer a hidden aid to individual answerers but a visible, platform-backed participant that competes directly with human answers for the same scarce resource, the limited attention and recognition of viewers.

Commented [DC1]: R1-1

This difference in visibility is not only substantive but also methodological. When external GenAI is used privately, its influence on human contribution is hidden and cannot be observed or measured, so the effect of an AI answer entering the competition is confounded with the unrecorded private use behind it. Internal GenAI removes this obstacle. Because the AI answer is labeled and because the platform decides which questions receive one, the entry of an AI competitor becomes a visible and well-defined event. This visibility is what allows us to isolate the effect of an AI answer on human contribution, an effect that remains obscured in the external GenAI setting.

This competitive arrangement raises questions that are central to the design and sustainability of online Q&A communities. Precisely because internal GenAI is the only lever a platform can pull, understanding its consequences for human contributors is not a peripheral design question but the decisive one. When a labeled, platform-endorsed, first-position AI answer competes directly with human answers, will human contributors withdraw their effort? Will they adjust what they write in order to distinguish their contributions from the AI answer? The answers matter for whether internal GenAI helps a community retain its members or accelerates the departure of its human contributors.

We study these questions in a setting where users retain access to external GenAI, and it is useful to be precise about what the platform does and does not restrict. The community in our setting discourages the posting of AI-generated answers, that is, answers that a user copies from an external tool and submits as the user's own work. It does not, and in practice cannot, prevent users from consulting ChatGPT or learning from it before they post. The baseline against which we measure the effect of internal GenAI is therefore one in which external GenAI remains available for private use and continues to divert demand. Our estimate is the effect of adding a visible, platform-integrated internal GenAI answer on top of this baseline.

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While these questions have important implications for the design and sustainability of online Q&A communities, the answers are characterized by theoretical tensions. The answers to our questions are not obvious, because the entry of a competing AI answer pulls human behavior in opposing directions. Based on Kahneman's dual-process theory (Evans & Stanovich 2013; Kahneman 2003, 2011) and competition dynamics theory (Chen et al. 2007; Chen, 2009), we organize these forces around two parts of the internal GenAI answer: its label and its content. The AI label signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one. The presence of a weak competitor can motivate human contribution rather than discourage it. The AI content works in the other direction. When the AI answer is long and appears comprehensive, potential answerers may judge that little of value remains to add, and their contribution falls. The same two-part logic applies to the quality of human answers, where a higher-quality AI answer sets a higher benchmark that human contributors respond to. The net effect therefore depends on how the label and the content combine.

To investigate internal GenAI's impacts in the presence of external GenAI availability, we leverage a randomized experiment conducted by a leading online technical Q&A community in China, where the platform introduced a GenAI bot to provide answers to randomly selected questions. This setting is particularly well-suited for our study due to the platform's early adoption of internal GenAI, its IT professional user base that faces a competitive environment where internal GenAI directly competes for limited recognition and user attention, and the randomized design that enables clean causal inference. Critically, our treatment effect should be interpreted as the impact of adding visible, platform-integrated internal GenAI answers on top of a baseline where users have access to external GenAI tools like ChatGPT.

Our results show that the effect of internal GenAI is not a single deterrence effect but the combination of two opposing forces. On answer supply, the AI label raises human contribution. This pattern fits the view that users treat the AI as a weak competitor whose presence invites participation. Longer AI content works against this, but not because length signals higher quality. A longer answer creates a perception of coverage: the answer appears comprehensive, so the room left to add something of value seems to shrink, and human contribution falls. Because AI answers in our setting typically carry substantial content, the average effect on supply is negative. On answer quality, the AI content again does the work. Higher-quality AI answers raise the quality of the first human answer, and they do so because human contributors respond by improving their answers' accuracy based on the AI answer. Separating the AI label effect from the AI content effect explains why the average effect can differ from either part on its own, and it provides a framework for reasoning about how AI will affect human contribution as its content characteristics change.

Our paper makes theoretical and practical contributions. We extend research on GenAI in online communities by studying platform-provided internal GenAI as a governance arrangement distinct from external GenAI, and by examining its effect on human contributors in a competitive environment where external GenAI remains available. Building on dual-process theory and the logic of competition dynamics, and by separating the AI label effect from the AI content effect, we show that the community's response to an AI competitor is neither uniformly negative nor uniformly positive and depends on identifiable features of the AI answer. This decomposition also lets us reason about future settings in which AI content continues to improve. For practice, our findings help platforms integrate AI while preserving the distinctive value of human contributions in a world where external AI tools are ubiquitous.

2. LITERATURE REVIEW

Our study draws on two literatures. The first concerns what drives a community member's decision to answer a question that already has prior contributions. The second concerns how generative AI has reshaped online Q&A communities. We organize the second literature by the structural role AI plays relative to the human answerer, because that role determines both what each study is able to identify and what it necessarily leaves open. We then draw on a third literature, on human responses to identified machine agents, which supplies the bridge between the two.

2.1. How Prior Contributions Shape Human Contribution in Online Q&A Communities

Our setting concerns how the presence of a prior AI contribution affects a subsequent human one in an online Q&A community. We therefore build on research that examines how existing content in a question thread influences human contribution. Contributing an answer is seldom an isolated act. When a potential contributor arrives at a question, that question usually already displays some prior content, and this content is the most immediate evidence about the value of adding one more answer.

Prior work suggests that existing content shapes subsequent answering through four related but analytically distinct features: presence, volume, quality, and source. The first two features concern whether a new answer is still needed. The mere presence of prior answers can reduce a potential contributor's perceived marginal value, because the question appears at least partially solved; this logic is consistent with classic free-riding arguments, in which individual incentives decline as others have already contributed (Olson Jr 1971; Zhang & Zhu 2011). Beyond presence, the amount of existing content further affects the perceived room for contribution. As answers accumulate, each additional answer tends to add less new information, and threads with more prior answers may therefore offer fewer opportunities for a newcomer to make a distinctive contribution (Zhang et al. 2022). The latter two features concern the evaluative standard created by prior content. High-quality answers do not simply substitute for later answers; they also signal the level of effort and expertise expected in the thread, thereby shaping the quality of subsequent contributions (Shi et al. 2024). Finally, the source of a prior answer provides a social cue about who has already spoken. A contributor's identity or standing can affect how newcomers assess their own relative position and whether their contribution is likely to be valued; consistent with this comparison logic, Chen et al. (2010) show that information about one's standing relative to peers changes subsequent contribution behavior. Together, this literature shows that prior contributions matter not only because they provide information, but also because they alter the perceived need, opportunity, benchmark, and social meaning of adding another answer.

A broader but separate body of work identifies other determinants of participation. These include question formulation and formatting (Burke et al. 2010; Calefato et al. 2018; Li et al. 2023), monetary and non-monetary incentives whose effects depend on context (Berger et al. 2016; Jiang et al. 2024; Wang et al. 2022), feedback and reciprocity (Li et al. 2020; Moon & Sproull 2008), and gamification (Chen et al. 2022; Meng et al. 2013). These determinants are properties of the question or the platform, not properties of the prior content that a newcomer confronts upon arrival. They are also fixed before any particular competitor appears. Our study holds them constant by design.

Across this literature, the prior contributor is always another human. Whether presence, volume, quality, and source carry the same meaning when the prior contributor is an AI is not settled by existing evidence. There is good reason to expect that they do not, and this is the question our setting allows us to address.

2.2. The Impact of GenAI on Online Q&A Communities

A large and fast-growing literature documents the consequences of generative AI for online Q&A communities. Table 1 summarizes this work. Rather than group studies by platform or by outcome, we group them by the structural role AI occupies, because the role determines the causal channel a study can isolate. Two dimensions define this role. The first is ***where AI acts***: on the flow of questions into the community, or on the production of answers within it. The second applies once AI acts within the answering process, and concerns ***what AI is to the human answerer***: an instrument the answerer uses while retaining authorship, or a rival that competes with the answerer under its own explicit label. These dimensions yield three roles. AI can act on the incoming questions and thereby substitute for the community itself; within the answering process, it can serve as the answerer's instrument, or it can enter as an identified rival contributor. The first two roles have been studied entirely in the context of external GenAI, tools that operate independently of any platform. The third role, which our study examines, is occupied by internal GenAI, and as we develop in Section 2.2.3, it constitutes a distinct phenomenon rather than external GenAI relocated inside the platform.

2.2.1. AI as a Substitute for the Community

The first and largest stream studies AI operating *before* questions reach the community. General-purpose tools such as ChatGPT answer questions directly, and the consistent finding across platforms and time windows is a decline in question supply (Burtch et al. 2024; Quinn & Gutt 2025; del Rio-Chanona et al. 2024; Sanatizadeh et al. 2025; Xue et al. 2026). A notable exception is Rao et al. (2026), who find that users who actively employed ChatGPT to generate answers subsequently posted more questions. These questions were also longer and more novel, and attracted more upvotes. This pattern suggests a complementary effect at the individual user level, one that contrasts with the aggregate decline documented in studies examining platform-wide trends. The overall reduction in question volume stems primarily from the fact that simpler questions are now answered by ChatGPT. Consequently, the remaining questions tend to be more complex and higher-quality (Quinn & Gutt 2025; Sanatizadeh et al. 2025; Xue et al. 2026). In this stream AI is a rival to the *platform*: it competes for the question itself and does not appear inside the community.

This stream establishes an important empirical baseline, and Xue et al. (2026) show that the decline is sustained rather than transitory. By construction, however, it speaks to whether questions arrive, not to how answerers behave once a question has arrived. It also describes an environment that platforms cannot control, since users cannot be prevented from consulting external tools. The managerially tractable question, that is, what a platform should do about the questions that do arrive, lies outside its scope.

2.2.2. AI as the Answerer's Instrument

A second stream studies AI as an *input* to human answers. Here AI lowers the cost of drafting a response, and the human retains authorship. The relationship is collaborative rather than competitive: human and AI work together in a single answering process. Several studies find that this raises answer supply (Fang & Zhou 2026; Shan & Qiu 2026), and evidence from the reverse direction points the same way, as bans on AI-generated content reduce both answer volume and acceptance rates (Wang & Zheng 2024). Other studies report declines in the volume of qualified contributions (Li & Kim 2025; Sanatizadeh et al. 2025), so the evidence is not uniform.

Findings on answer style are similarly divided, with Sanatizadeh et al. (2025) reporting longer and less readable answers and others shorter and more readable ones (Borwankar et al. 2026; Fang & Zhou 2026; Shan & Qiu 2026).

Part of this divergence might reflect differences in platform, unit of analysis, and how style is operationalized. Taken together, these studies establish an important and non-obvious result: AI adoption changes not only how much answerers write but how they write. We suggest that the divergence is consistent with AI exerting two forces that this stream's designs cannot separate. As a drafting aid, AI encourages elaboration and length. As a source of competitive pressure on the community as a whole, it pushes answerers toward concise, differentiated contributions that AI cannot readily replicate. Which force dominates should depend on which role AI plays in a given setting, and this stream cannot answer the question because AI's contribution is not visible as such.

That invisibility is the stream's defining limitation. A human answerer's use of AI cannot be observed by other community members, and in most cases cannot be observed by the researcher either. Studies in this stream can therefore detect that answering behavior changed, but cannot determine whether the change was produced by AI-generated *content* or by the *knowledge* that a contribution is AI-generated. Nor can they observe the competitive signal an answerer faced at the moment of deciding, since no answerer in this stream ever sees a contribution identified as machine-produced.

2.2.3. AI as an Identified Rival Contributor

A third and much smaller stream examines AI that posts answers under its own explicit label, alongside human contributors, on the same questions. Here AI is a rival to the *human* *answerer*, competing for the same viewers' attention and recognition, and both the source and the content of its contribution are visible.

Only Su et al. (2025) has examined this configuration empirically. Studying Quora's deployment of Sage, they report positive effects on human contribution, including increases in the number and length of answers. Two features of their setting differ from ours in ways that plausibly determine the sign of the effect. First, *timing*: their AI answers are added roughly three months after the original post, by which point questions have already accumulated human responses, so AI enters as a retrospective supplement to an established discussion rather than as a competitor whom human contributors encounter on arrival. Second, *domain*: Quora is a general-interest platform spanning philosophy, personal advice, and cultural discussion, where the value of a human contribution rests substantially on subjective interpretation and lived experience that AI cannot easily substitute for. In technical communities, contributions are evaluated primarily on correctness and efficiency, precisely the dimensions on which AI capability is strongest, and a labeled AI answer that arrives within minutes competes directly with human answerers from the outset.

Whether internal AI complements or displaces human contribution is therefore likely contingent on domain and on timing. The configuration most consequential for platform design, immediate AI competition in a domain where AI is capable, remains unexamined. We return to reconciling our findings with those of Su et al. (2025) in Section 7.

2.3. Human Responses to Identified Machine Agents

Because internal GenAI is explicitly labeled, our setting activates a literature on how people respond to machine agents they know to be machines. This work establishes that disclosure alone changes behavior. People discount algorithmic advice after observing it err, even when the algorithm outperforms humans (Dietvorst et al. 2015), while over-weighting algorithmic judgment relative to human judgment in other conditions (Logg et al. 2019). The direction of the effect depends on how subjective or human-dependent the task is perceived to be (Castelo et al. 2019), which implies that the meaning of an AI label is domain-specific rather than fixed. Related literature shows that responses to capable machines are not confined to substitution or withdrawal: exposure to AI has been shown to increase the novelty of subsequent human decisions (Shin et al. 2023), and studies of AI-exposed labor markets document both displacement and adaptive repositioning toward tasks where human value is distinctive (Demirci et al. 2024; Hui et al. 2024; Yiu et al. 2026).

Two implications follow for online Q&A communities. First, an AI label may carry motivational content in its own right, independent of what the AI actually wrote. Second, a capable AI rival may induce humans' motivation to excel, and it may raise rather than lower the quality of human contributions. Neither implication is testable where AI's presence is unobservable to the contributor, which is the case throughout Sections 2.2.1 and 2.2.2. Both become testable when the platform itself deploys a labeled AI answerer.

2.4. Positioning and Contribution

The first two streams in Section 2.2 studying external GenAI rest on a shared premise that matters for theory development. In the first stream, AI never enters the answering process. In the second, it enters invisibly. In both, the effect of AI content and the effect of the AI label are bundled, and the competitive signal the answerer confronted is unobserved. Platform-deployed, explicitly labeled AI dissolves this bundle. Because both the source and the content of the AI contribution are visible to community members and to the researcher, it becomes possible to estimate the effect of the AI *label* separately from the effects of specific content attributes, namely length and quality, and to test the distinct decision processes through which each operates.

Table 2 positions our study against the three most closely related studies on the dimensions that drive this identification.

Three features of our setting make this decomposition possible and make its results consequential. First, the AI is deployed by the platform under direct governance, so its presence is a design choice rather than an uncontrollable environmental shift, which places the findings within managerial reach. Second, assignment of AI answers is randomized at the question level, which permits clean causal inference and allows us to compare questions whose earliest answer is AI-generated with otherwise identical questions whose earliest answer is human-generated. Third, the AI responds almost immediately, before any human contributor arrives, in a technical domain where its capabilities are strongest, which places human contributors in the most competitively demanding configuration a platform is likely to create. Because external tools remain freely available throughout our observation window, our estimates should be interpreted as the effect of *adding* visible, platform-integrated AI on top of a baseline in which external AI is already accessible. This is the decision platforms actually face: they cannot eliminate external GenAI, but they can choose whether and how to deploy their own.

Understanding its effects is therefore directly relevant to platform design and community management. Our study addresses this gap by examining how platform-deployed internal GenAI (on top of existing external GenAI availability) affects human answer supply and contribution quality in an online Q&A community setting.

Table 1. Summary of Empirical Research on the Effects of Generative AI on Online Q&A Communities

Study	Platform (domain)	Source of variation	Unit of analysis	AI source identifiable to users	Question supply	Answer supply
Role 1. AI as a substitute for the community (AI acts before questions reach the platform)						
Xue et al. (2026, ISR)	Stack Overflow (technical)	ChatGPT release	Topic-day	No	–; remaining questions higher quality	
Quinn & Gutt (2025, JMIS)	Stack Exchange (technical)	ChatGPT release	Topic-week	No	–	
Sanatizadeh et al. (2025, I&M)	Stack Exchange (technical)	ChatGPT release	Post-day	No	–; remaining questions more complex	–; longer, less readable
Rao et al. (2026)	Stack Overflow (technical)	ChatGPT release	User-week	No	+ (individual level); remaining questions longer & more novel	
Burtch et al. (2024)	Stack Overflow (technical)	ChatGPT release	Topic-week	No	–	
del Rio-Chanona et al. (2024)	Stack Overflow (technical)	ChatGPT release	Topic-week	No	–	
Role 2. AI as the answerer's instrument (AI enters human answers as an unattributed input)						
Shan & Qiu (2026, ISR)	Stack Overflow (technical)	ChatGPT release	User-day	No		+; shorter, more readable
Fang & Zhou (2026, I&M)	Zhihu (general)	ChatGPT release	User-month	No		+; shorter
Borwankar et al. (2026)	Stack Overflow (technical)	ChatGPT **ban**	Answer-user-day	No		Answers shorter & more negative
Li & Kim (2025)	Stack Overflow (technical)	ChatGPT release	User-day	No		–
Wang & Zheng (2024)	Stack Exchange (technical)	ChatGPT **ban**	Topic-week	No	+	+; acceptance +
Role 3. AI as an identified rival contributor (AI posts labeled answers alongside humans)						
Su et al. (2025)	Quora (general)	Internal GenAI; Answers added ≈3 months after posting	Question-week	Yes (labeled)		+
This study	SegmentFault (technical)	Internal GenAI; **Randomized exp. at question level**; AI answers generated immediately after question posting and listed before any human answer	Question	Yes (labeled)	Not applicable (due to question level randomization)	–

Notes. 1. Signs ("+"/"–") denote the direction of each study's estimated effect; blank cells indicate outcomes not reported here by design.
2. Sign convention for ban-based studies. Most studies identify effects from the release of ChatGPT, so their signs directly capture the effect of AI becoming available. Two studies instead exploit a ChatGPT ban (Wang & Zheng 2024; Borwankar et al. 2026). Because a ban is the withdrawal of AI, the raw estimates in these studies measure the effect of removing AI and are the mirror image of the release-based estimates. To keep all rows comparable, we reverse their signs. For example, Wang & Zheng (2024) find that banning ChatGPT decreases question supply, answer supply, and answer acceptance; equivalently, the availability of ChatGPT increases these outcomes, which is why this row is recorded as "+" for question and answer supply.
3. Su et al. (2025). This is an internal-GenAI setting on Quora. The questions were created before the observation panel began, and platform AI answers were introduced only at the policy launch (≈ Jan 2023). Following their design, the sample keeps "only those questions that were created before the start of our panel data collection (i.e., November 2022)," and treated pages receive AI answers when "AI answers are presented on treated question pages" at launch. The AI answers therefore appear roughly three months after the questions were originally posted.

Table 2. Positioning Relative to the Three Most Closely Related Studies

Dimension	Xue et al. (2026)	Shan & Qiu (2026)	Su et al. (2025)	This study
Locus of AI	External	External	Platform-internal	Platform-internal
AI's relation to the human answerer	None; diverts questions before arrival	Collaborator and writing aid	Delayed supplement to an existing discussion	Direct rival on the same question
AI content identifiable as machine-produced	No	No	Yes (labeled)	Yes (labeled)
Timing of AI answer relative to human answers	No	No	≈3 months after posting, once human answers have accumulated	AI answers are generated immediately after question posting and listed before any human answer
Domain and basis of contribution value	Technical; correctness	Technical; correctness	General-interest; experience and interpretation	Technical; correctness and efficiency
Identification strategy	Natural experiment	Natural experiment	Natural experiment and lab experiment	Field randomization at the question level
Separates AI *label* from AI *content*	No	No	Not the focus	**Yes**; central to the design
Primary outcome	Question supply	Answer supply and style	Answer supply	Answer supply **and** contribution quality

3. THEORETICAL DEVELOPMENT

3.1. The Encounter as a Competitive Episode

Consider the following situation. A member of a technical Q&A platform posts a question about code that fails. Within seconds, the platform attaches an answer generated by its own AI system, marked with a label that identifies the author as a machine. A human member arrives at the thread later, sees that a machine has already responded, and must decide two things: whether to write an answer at all, and, if so, how much effort to put into it.

We treat this encounter as a *competitive episode*, and develop a theory of how the member sizes up the machine rival and how that assessment shapes what the member does next. Three considerations justify the choice of the competition lens.

First, the lens fits the situation the platform has created. The AI answer sits in the same thread and on the same screen as any human answer that follows, so a later human answer has to be worth reading in the presence of the AI answer.

Second, the lens tells us where to look. Research on competitive interaction holds that how an actor responds to a rival depends on how the actor appraises that rival, and that this appraisal is built from observable features: who the rival is, how capable it appears, and how much effort it appears to have put in (Chen et al. 2007; Chen, 2009).

Third, the competition lens generates richer predictions than a simple substitution account does. Prior work that treats generative AI as a substitute for human labor has established something important: these systems can perform much of the routine work human contributors once did. But that account can predict only decline. If the AI answer does work the member would otherwise have done, the member has less left to do and writes less. The substitution account contains no mechanism by which an AI answer could lead a member to write *more*, or to write *better*.

3.2. The AI Answer as a Rival with a Profile

As Section 2 reviews, existing studies ask whether AI raises or lowers human contribution. This is a reasonable starting point, but it treats “an AI answer is present” as a single condition with a single consequence. If the member’s response instead depends on how the rival is appraised, and if that appraisal is built from observable features of the rival’s answer, then two AI answers with different features are two different competitive situations and should produce different responses. In other words, the AI answer is a rival with a *profile*.

Our central claim is that the effect of an AI answer depends on this profile, and that the dependence differs from one feature to another. As AI answers get longer, and as they get better, the consequence of having one beneath the question changes, and it changes in different directions for different features.

The profile has three features. The *label* indicating machine authorship is present; it does not vary in degree, so it produces a *level* effect. *Length* and *quality*, by contrast, vary continuously from one answer to the next, so they produce *slopes*: they determine how the consequence of an AI answer changes as its content changes. Therefore, our predictions take two forms, one about the level attributable to the label, and one about the direction of the slope on length and on quality.

Our theory addresses three questions. What does the label alone do when content is held constant? How does the consequence of an AI answer change as the answer gets longer? And how does it change as the answer gets better? Because a member facing a rival makes two decisions, whether to answer and how good the answer will be, these three questions combine to yield six predictions.

Two of the six slopes are predicted to be null. We recognize that a null prediction may look like a weakness. But a framework that predicts an effect in every cell cannot be wrong in any interesting way. If every feature were expected to move every outcome, almost any result would count as support, and the framework would say little more than "capable AI discourages people." By predicting that two specific cells are null, we commit in advance to results that would count against us. What distinguishes our account is not the claim that AI answers matter, but the specific pattern of which feature affects which decision, and that pattern follows from the cognitive process we describe next.

3.3. Two Decisions, Three Signals, and the Timing That Links Them

Competition research tells us what a contributor would *want* to know about a rival. It does not tell us which judgments the contributor can actually form, or when. For that, we draw on the two-systems account of thinking (Evans & Stanovich 2013; Kahneman 2003, 2011). System 1 operates automatically, demands almost no effort, and responds to surface cues. System 2 is deliberate and effortful, and it engages only when the situation appears to justify the cost. Kahneman (2011) calls this default the *law of least effort*.

Two decisions. The member first decides whether to answer at all; we call this the *entry decision*. Having decided to answer, the member then decides how good the answer will be; we call this the *quality decision*. Entry is a yes-or-no choice that can be settled quickly, before the AI answer has been read. The quality decision is a matter of degree, and it is where the member's time and knowledge are actually committed.

Three signals. The AI answer carries three pieces of information. The *source* (the label) tells the member what kind of rival this is. The *length* signals how much of the answerable ground the rival appears to have taken, and therefore how much room is left. The *benchmark* is the quality of the rival's answer, which signals the standard a contribution must meet to be worth reading.

Timing. These signals do not all arrive at once, and the timing is what ties each signal to a particular decision. Source and length are *glance signals*: the label is plainly visible and so is the physical size of the answer, and neither requires reading. The benchmark is different in kind. A member learns the quality of an AI answer only by reading it and checking its reasoning or its code against the problem the questioner posed. That is exactly the sort of deliberate work the law of least effort withholds.

Figure 2 presents this structure. The three signals run across the columns, arranged by whether they are available before reading or only after it, and the two decisions run down the rows. Each cell records the predicted sign, or a predicted null, for one signal-by-decision combination, and the six cells together correspond to the six hypotheses developed below.

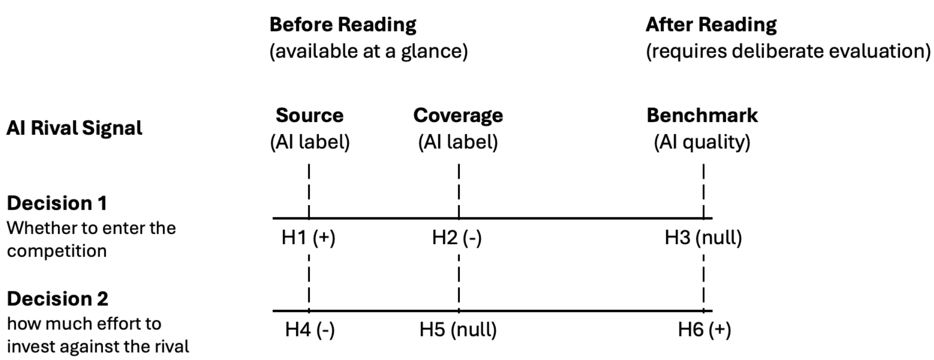


Figure 2. Conceptual Framework

3.4. The Entry Decision

3.4.1. Source: The Label Component

The source is the first signal registered, and it can shape the entry decision because it is available before any content is read. We argue that the label leads the member to appraise the AI as a *weak* rival, and that a weak rival encourages entry.

Noticing a simple visual marker is enough to know that the answer came from the platform's AI system. System 1 then judges the rival's capability through the *availability heuristic*, so the judgment rests on whatever beliefs about machine-generated answers come to mind most readily (Tversky & Kahneman 1973). Two beliefs are relevant.

The first belief is a general impression that generative text is bland and formulaic, a quality that popular discussion captures with the phrase "AI slop." The evidence is consistent with this impression. Readers and trained evaluators tend to rate machine-generated writing as generic and lacking a distinctive voice, and automated text is marked by repetitive, low-variability phrasing that makes it recognizable as machine-produced (Anderson et al. 2024; Doshi & Hauser 2024). The sense conveyed by "AI slop" is thus that the output follows a predictable template rather than saying anything sharp or particular.[2]

The second belief is specific to technical Q&A, and it is the one our argument rests on. Members of technical communities have been openly skeptical of AI-generated answers, and their skepticism has an empirical basis. Kabir et al. (2024) report that more than half of

[2] We should be precise about the belief we attribute to this population, because it is narrower than the general "AI slop" view. Members of SegmentFault are professional developers, and many had already adopted general-purpose AI tools for routine tasks, so they did not hold the blanket view that AI writes poorly. Their reservation was domain-specific. Throughout our observation window, developer discourse continued to describe AI-generated answers as unreliable on technical problems, and members had not revised that assessment in light of large language models. The belief, in short, is that the AI has probably not solved the particular problem at hand.

ChatGPT's answers to Stack Overflow questions contain incorrect information, that these answers are considerably more verbose than human answers, and that readers miss the errors in a substantial share of cases. Stack Overflow banned AI-generated answers for this reason. Disclosure compounds the effect, since information is judged less accurate when it is identified as machine-generated (Longoni et al. 2022). The consequence is not just doubt but cost: an answer that looks plausible but may be wrong has to be verified before it can be used, and verifying it means working through the reasoning oneself. Therefore, an AI answer does not close a question the way a trusted human answer would.

Appraising the AI as a weak competitor pushes toward entry. The AI answer sits under the same question, faces the same readers, and competes for the same attention, precisely the condition under which an actor is most likely to notice a rival and respond (Chen et al. 2007; Chen 2009). The mere presence of a competitor in the same thread is an invitation to respond, and a *weak* competitor makes responding attractive: the risk of losing is low, and a competent developer can supply the correct answer the AI failed to supply.

We recognize that this runs against the intuition that AI crowds out human contribution, and some skepticism toward a prediction in which adding a competitor *raises* participation is appropriate. But our claim is narrow. It concerns the label alone, with content held constant, that is, what a marker of machine authorship does when it says nothing about the substance of the answer.

H1. Holding the content of the AI answer constant, the AI label increases human answer supply.

3.4.2. Length: The Coverage Slope

The second glance signal is length, and it pushes entry in the opposite direction. Length is processed through *attribute substitution*, in which System 1 replaces a difficult judgment with an easier one associated with it (Kahneman 2011; Kahneman & Frederick 2002). The difficult judgment is how much of the question the rival has covered; the easy substitute is physical size. A visibly long AI answer is read as the work of a competitor that has treated the question comprehensively, and this impression revises the weak-rival appraisal that the label alone produced.

Our setting makes this substitution reliable. Technical answers are broken into code blocks, numbered steps, configuration snippets, and shell commands, so their structure is visible without reading, and a long answer in this format appears to list several possible causes or several ways to fix the problem. Thus, length is a credible indication of how much ground has been taken.

What matters at entry is how much room is *left*. The more of the question the existing answer appears to cover, the less room appears to remain, the smaller the margin a new contribution can achieve, and the weaker the incentive to enter. Two points deserve emphasis. First, this is a claim about *coverage*, not quality: the member has not checked whether the long answer is correct, and an answer can be comprehensive and wrong at the same time. Second, H1 and H2 act on the same decision in opposite directions, and both rest on glance signals. This is why the consequence of adding an AI answer cannot be a single number. It is positive where AI content is thin and negative where it is thick, and the point at which it switches sign depends on the answers a given platform actually posts.

H2. The effect of an AI answer on human answer supply becomes more negative as the length of the AI answer increases.

3.4.3. AI Answer Quality: Why It May Not Affect Entry

The label and the length can both be seen at a glance; one is a tag, the other a shape on the screen. Quality becomes available only afterward, because it requires reading the answer through and checking it against the specific problem. Therefore, entry is shaped mainly by what arrives first.

This argument rests on the division of labor between the two systems. System 1 forms an appraisal from the cues at hand and returns a disposition to act or to pass; System 2 is then in a position to examine that disposition and either endorse it or overturn it. The dual-process account emphasizes that System 2 normally *endorses*, because it is recruited only when the fast route fails to deliver a confident answer, and not recruited when the fast route delivers one cheaply, which is the law of least effort at work (Kahneman 2011). Here the fast route does deliver a confident answer: the label and the length together produce a coherent impression of how much ground the rival has taken, so nothing calls for the more expensive judgment that assessing quality would require.

Even a member who does read the AI answer tends to confirm the first impression. An answer looks comprehensive because of how it is laid out, but correctness depends on whether the code runs, whether the named libraries and versions are the ones actually in use, and whether the diagnosis fits the setup described. Checking these things means working through the reasoning, which is demanding even for skilled programmers (Lee & Lee 2025), and errors in AI answers often go undetected for exactly this reason (Kabir et al. 2024). Careful reading is thus costly to start and unreliable once started, so it overturns the first impression less often than one might expect.

Reversals do occur in both directions. A member who meant to pass may find the AI answer mistaken and write up their own answer after all; a member who meant to write up their own answer may find the AI answer sound and stop contributing their own answer. We do not claim that either pattern is impossible. We claim that neither is systematic enough to produce a detectable relationship between the quality of an AI answer and the rate at which human answers appear.

We should also be clear about what H3 does *not* say. It does not say that members cannot assess the quality of an AI answer, and it does not say that quality is irrelevant to what they do. Section 3.5.3 predicts that quality matters a great deal, precisely because the quality decision is made *after* the member has committed to answering, at which point reading the AI answer is no longer optional.

H3. The effect of an AI answer on human answer supply does not vary appreciably with the quality of the AI answer.

3.5. The Quality Decision

Once a member has decided to contribute, the question becomes how good the answer needs to be. That judgment forms in two stages, first from the appraisal made before reading and then from what reading reveals, so H4 and H6 describe two stages of a single process.

3.5.1. Source: The Label Component

The member has already appraised the AI as unlikely to have solved the problem, and that appraisal does not vanish once writing begins. Having judged the rival to be weak, the member expects that even a fairly modest answer will be enough to come out ahead. The law of least effort holds that people supply effort only up to the level a goal requires, and no further (Kahneman 2011); here, the appraisal has set the goal low. Competition research points the same way: the intensity of a competitive response depends on how much the rival appears to have invested, and weak performance by a rival reduces the resources an actor commits (Chen, 2009; Hsieh et al. 2015). Facing a rival that appears incapable, the member is highly confident of winning and invests only what winning requires.

"Doing just enough" takes a concrete form here. A technical answer can be reduced to a working repair, such as a corrected line of code, the name of the relevant configuration flag, or a link to the documentation section that resolves the issue. Such an answer leaves out *why* the repair works, *when* it fails, *what* the alternatives are, and *what* to watch for. Two features of the setting make this minimal response likely. First, a short answer is viable, because a technical question is settled by a solution that works, so a brief answer that fixes the problem can beat a longer one that does not. Second, writing more is expensive: a thorough answer requires reconstructing the questioner's situation and often writing and testing code, whereas the minimal repair can be supplied from knowledge the member already holds.

H4. Holding the content of the AI answer constant, the AI label decreases the quality of the human answer.

3.5.2. Length: A Cue Replaced by Content

Length plays no major role in the quality decision, and the reason lies in which system is operating when that decision is made. Entry is settled quickly, on cues visible before the answer is read, and length is one of them. The quality decision is settled afterward, by a member who has committed to answering and has therefore read the AI answer through. Such a member has the *content* of the rival answer in front of them, and there is no reason to consult its length when its substance is available.

The beliefs of this population push in the same direction. Members associate AI answers with padding, and the association has a basis: AI answers to technical questions are substantially longer than human answers while being incorrect in more than half of cases (Kabir et al. 2024). A member who holds this belief has reason to treat length as an indication of how much has been *attempted* rather than how well it has been *done*, so deliberation tends to discount the cue rather than defer to it. We would not claim that length plays no role at all in the quality decision, since a member may retain some residual impression of the rival's scope from the initial scan. We claim only that this residue is weak relative to what reading the answer supplies, too weak to generate a systematic relationship between the length of an AI answer and the quality of the human answers that follow.

This is why the same signal is decisive for one decision and marginal for the other, and it is the second point at which our framework yields a prediction that a general capability-deterrence account cannot. The boundary of the claim should also be stated. H5 does not imply that long AI answers are equivalent to short ones in every respect; H2 predicts that they are not. It implies only that once a member is engaged closely enough to weigh the standard their answer must meet, length carries little information about what that standard is.

H5. The effect of an AI answer on human answer quality does not vary with the length of the AI answer.

3.5.3. Benchmark: Calibrating to a Demonstrated Standard

A member who has decided to contribute still has to judge how good the answer needs to be. Questions on SegmentFault are usually brief, often little more than a pasted error and a request for help, and they rarely state what would count as an adequate reply, whether that is a corrected line of code, an explanation of the cause, or a working alternative. The member is aiming at a target that has not been defined.

The AI answer defines it, because it is a *completed instance* of the task. It shows how much of the problem a reply is expected to cover, how closely it should engage the questioner's setup, and whether the cause needs explaining or only the fix. What was an open question becomes a comparison against a visible reference level, which we call the *demonstrated standard*.

How high that standard sits depends on how good the AI answer is. Suppose a member can write two sentences explaining the actual cause of the error. If the AI answer is weak, those two sentences are already the better contribution, and the member has no reason to write more. If the AI answer is strong, explaining the mechanism and supplying code that runs on the questioner's setup, the same two sentences now look thin beside it. Therefore, a better AI answer sets a higher standard, and the member works toward a higher level in response.

The member can reach that higher level in two ways. One is to build on the AI answer, which is a "draft" already in hand. A good draft provides claims to check against the questioner's setup, a diagnosis to confirm or reject, and code to correct, so what the member writes keeps what the AI answer got right and fixes what it got wrong. The other is to add what the internal GenAI cannot supply, such as experience from the member's own deployments or opinionated insights about which approach to prefer. Either way, the prediction is the same: a better AI answer sets a higher standard, which pushes the member's answer toward higher quality.

H6. The effect of an AI answer on human answer quality becomes more positive as the quality of the AI answer increases.

3.6. The Predicted Pattern

Table 3 assembles the six predictions.

Table 3. Predicted structure of the response to an AI answer

Signal from the AI answer	Available before reading?	Effect on answer supply (entry decision)	Effect on answer quality (quality decision)
Source, the AI label (level)	Yes	Positive (H1)	Negative (H4)
Length, coverage (slope)	Yes	Negative (H2)	Null (H5)
Quality, the benchmark (slope)	No	Null (H3)	Positive (H6)

The first row reports the *level* component, the effect of the label with content held constant, and its two entries carry opposite signs. The second and third rows report the *slopes* on the content features, and each carries one signed entry and one negligible entry. Read as a whole, the table shows a crossing pattern: length moves supply but not quality, quality moves quality but not supply, and the label enters both outcomes in opposite directions.

4. RESEARCH CONTEXT

4.1. SegmentFault and a Randomized Field Experiment on Internal GenAI

Our study examines data from SegmentFault, a prominent Chinese technical Q&A community specializing in IT and software development. Founded in 2012, SegmentFault serves nearly 7 million users and facilitates knowledge exchange across web/mobile development, AI, cloud computing, and cybersecurity. As of April 2024, the platform contains over 310,000 technical questions with 1.9 answers per question on average. Among global technical Q&A communities, SegmentFault ranks sixth in questions asked, third in answers provided, and first in answer-to-question ratio.[3] It constitutes an essential resource for Chinese developers, programmers, and IT engineers who seek knowledge, solutions, and collaboration opportunities.

SegmentFault's ecosystem relies on user-generated content and interactions. Members pose questions, provide answers, participate in discussions, and express approval through "accept" or "like" actions. The operations team reviews user-contributed solutions before publishing them. Features enhancing engagement include comment sections for each answer, a voting system enabling questioners to "accept" answers while viewers upvote or downvote, and a reputation system rewarding members with points and medals based on contribution quantity and quality. Our study leverages a randomized field experiment conducted by SegmentFault.[4] Beginning September 13, 2023, the platform conducted an A/B test of its internal GenAI. Each newly

[3] This information can be found at: https://stackexchange.com/sites?view=list#technology-questions.
[4] The randomized experiment announcement can be accessed at: https://meta.segmentfault.com/questions/1010000053637863.

posted question was randomly assigned to either the treatment group (receiving an AI-generated answer) or the control group (receiving no AI answer). Each question received an independent random assignment upon posting, with no systematic variation by questioner characteristics, time periods, or other factors.[5] This question-level randomization offers a unique opportunity to investigate how internal GenAI affects user behaviors and community dynamics in the presence of external GenAI. Online Appendix A provides detailed background information.[6]

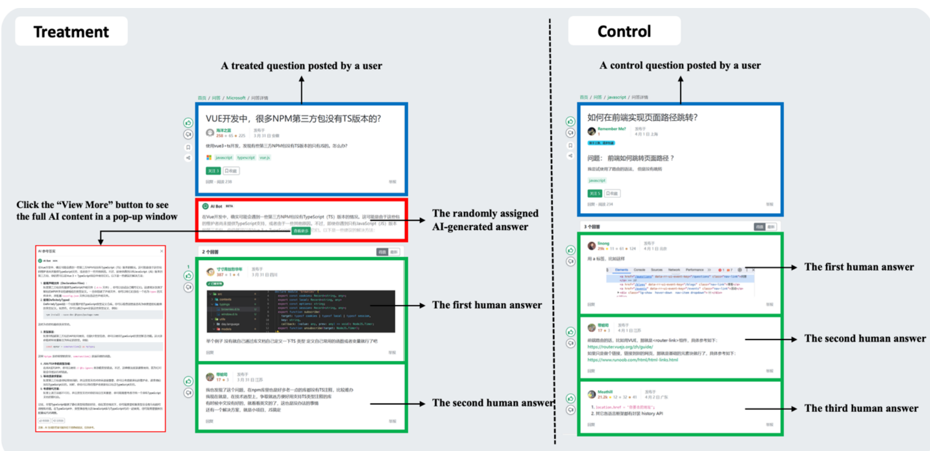


Figure 3. Illustration of treatment and control questions

Figure 3 illustrates differences between "treated questions" (with AI answers) and "control questions" (without). While the platform records precise timestamps for human answers, it omits this for AI answers. We observe that AI answers appear within five minutes of question posting, substantially faster than human answers, which arrive after an average nine-hour delay. Consequently, AI answers typically occupy the first position for treated questions. AI answers also differ from human answers because they cannot receive endorsements from

[5] We confirmed with the SegmentFault administrators who oversaw and implemented the randomized experiment that the random assignment did occur at the question level.
[6] The Online Appendix is available at: https://researchbox.org/5255&PEER_REVIEW_passcode=UFACJZ.

questioners or votes from viewers.

Community members can provide answers regardless of the presence of an AI-generated answer. When an AI-generated answer exists, viewers typically see this content before human-generated answers. The platform initially displays only two lines of AI-generated content on the question page. Viewers can access the complete answer via a “View More” button, which triggers a pop-up window labeled “AI Answers” with a disclaimer that alerts users to potential errors or inaccuracies and indicates the content is for reference only.

Two features of the deployment matter for the analysis that follows. The AI answer is explicitly labeled, so the source signal in our framework is observable to community members and to us. And the AI answer arrives fast, well before the average nine-hour delay of the first human answer, so it occupies the first position and is the rival a human answerer confronts on arrival rather than a supplement added to an existing discussion.

4.2. Data Collection

We crawled data from SegmentFault in February 2024. Our main analyses focus on the 150-day window from September 2023 (when the field experiment began) to February 2024. We collected all questions posted during this timeframe. Our dataset comprises 3,613 questions posted by 1,404 unique users during this 150-day window, averaging 24 new questions daily. AI-generated answers appeared in 1,766 questions (48.9% of total). We also collect other posts when necessary for mechanism analysis or posthoc analysis.

The platform assigns each question a tag as a content summary. Based on 384 unique tags, we classify questions into ten categories: web and mobile development, software design and architecture, cloud computing, databases, computer networks and systems, AI and machine learning, development tools and environment, programming languages, career and professional development, and other. "Web and mobile development" is the largest category, accounting for 57.5% of questions.

4.3. Summary Statistics

Table 4 reports summary statistics of our data. Questions in our sample average 86.809 Chinese characters and receive an average of 1.293 human answers, with 76.1% of questions receiving at least one answer. Questioners "accept" answers for 32.2% of all questions.

The treatment group exhibits notable differences from the control group. It receives fewer human answers (1.209 vs. 1.372) and has a lower proportion of accepted answers (29.3% vs. 35.0%). On question pages, viewers can upvote or downvote answers, with the page displaying net likes (upvotes minus downvotes) for each answer. Answers in our sample average 0.489 net likes, with the treatment group receiving fewer (0.436 vs. 0.536).

SegmentFault employs a reputation system where users accumulate reputation scores based on their contributions. Reputation scores in our sample range from -3 to 28,800, with an average of 378.337.[7] This average roughly corresponds to 10 questions, each receiving about 2 upvotes.

Table 4. Descriptive statistics of main variables

Variable	Mean and SD			Min	Max
	All samples	Treatment group	Control group		
question length (number of Chinese characters)	86.809 (113.981)	84.233 (89.347)	89.272 (133.323)	0.000	2702.000
number of human-generated answers received by a question	1.293 (1.166)	1.209 (1.133)	1.372 (1.192)	0.000	8.000
percentage of questions that received at least one answer	0.761 (0.427)	0.724 (0.447)	0.795 (0.404)	0.000	1.000
percentage of questions that have an answer "accepted" by the questioner	0.322 (0.467)	0.293 (0.455)	0.350 (0.477)	0.000	1.000
average number of net likes received by all the answers to a question	0.489 (1.096)	0.436 (1.041)	0.536 (1.140)	-3.000	14.000
reputation scores of the questioner at the time the question was posted	378.337 (1135.056)	334.484 (695.343)	420.267 (1433.522)	-3.000	28800.000

[7] In SegmentFault, receiving downvotes on an answer can result in deductions to the answerer's reputation score, potentially pushing it into negative territory.

Notes: 1. Columns 2-4 present the means of the main variables, with the standard deviations shown in parentheses below each mean.
2. The minimum question length of 0 represents questions where users posted content solely through images without accompanying text. These cases comprise 2.71% of our sample (98 out of 3,613 observations). Our main results remain unchanged when these observations are excluded.

4.4. Verifying Random Assignment

Our empirical strategy depends critically on the random assignment of questions to treatment and control groups. To validate this assignment, we conduct t-tests and examine standardized differences (Austin, 2009) across pre-assignment variables including questioner characteristics (reputation scores and gold, silver, bronze medal counts, total questions asked, total answers provided, account age), questioner platform activity in the 1, 2, and 3 months prior to the experiment (number of accepted answers, answers provided, questions asked, and comments posted), and question content features (length measured by Chinese character count, presence of multimedia elements such as photos, hyperlinks, tables, and block quotes). We also employ a large language model (LLM) to evaluate question characteristics, including linguistic complexity, technical jargon use, and difficulty (Li et al. 2024). Details about measure generation appear in Online Appendix B. We also account for question categories and the timing of question posting.

Online Appendix Table A1 presents balance check results demonstrating excellent balance across groups. All t-tests yield statistically insignificant results (p-values > 0.05), indicating no significant differences in pre-treatment variables. All standardized differences fall well below the 0.10 threshold, with absolute values ranging from 0.003 to 0.061.

Additionally, regression-based balance checks in Online Appendix Table A2 provide both an omnibus test of overall balance and coefficient-level diagnostics for individual covariates. This analysis accounts for question categories and posting timing. The results consistently confirm successful random assignment.

5. EMPIRICAL ANALYSIS

5.1. Main Variables of Interest

Answer supply. Our first outcome is the volume of answers a question attracts. We measure it as the logarithm of the number of answers a question receives within seven days of being posted, denoted Log(answer number in 7 days).

Answer quality. Our second outcome is the quality of human contributions. We employ an LLM to generate the quality measure. The detailed generating procedure and its validity assessment are provided in Online Appendix B. When studying human contribution quality, we apply this measure to the first human answer, denoted First answer quality (LLM).

The AI answer's profile. For treated questions, we characterize the AI answer along the two content dimensions the framework identifies. *Length* is the logged character count of the AI answer, and it stands for the coverage signal of Section 3.4.2. *Quality* is the LLM-generated quality described above, applied to the AI answer, and it stands for the benchmark signal of Section 3.4.3. Both are defined only where an AI answer exists and are set to zero for control questions.

5.2. Econometric Models

Our baseline model regresses each outcome on the AI treatment and, where relevant, its interactions with the moderators:

$$Y_i = \beta_0 + \beta_1 AI_i + \beta_2 (AI_i \times Length_i) + \beta_3 (AI_i \times Quality_i) + \gamma X_i + \delta_t + \varepsilon_i$$

where *i* indexes questions, Y_i is one of our two outcomes: the logarithm of the number of human answers received by question i within seven days, or the quality of the first human answer to question i. AI_i indicates treatment assignment (1 if AI-generated answer provided, 0 otherwise), X_i includes control variables (questioner characteristics, question content, and question type), and δ_t captures question posting date fixed effects.

The coefficients correspond one-to-one to the predictions in Table 3, and it is worth stating what each one says in words. β_1 is what an AI answer does when both content dimensions are at zero. The effect of an announced AI presence stripped of what the AI said. It is the test of H1 and H4. β_2 and β_3 show how the consequence of having an AI answer under a question changes as that answer's content changes. A negative β_2 means that a longer AI answer depresses human contribution more than a shorter one; an β_3 indistinguishable from zero means the consequence does not shift as the AI answer gets better. These are the tests of H2, H3, H5, and H6.

6. RESULTS

6.1. How the Effect on Answer Supply Varies with the AI Answer

Table 5 presents our main findings on internal GenAI-generated answers' impact. Columns 1 and 2 concern answer supply.

Table 5 reports the main results. Column 1 shows that the average effect of adding an internal GenAI on answer supply is negative and significant (-0.046, p<0.001). We further study how the effect varies with AI answer length and AI answer quality in Column 2.

Table 5. The impact of AI-generated answers on answer supply and answer quality

	(1)	(2)	(3)	(4)
DV	Log(answer number in 7 days)		First answer quality	
AI_i	-0.046***	0.188*	0.072	0.411
	(0.011)	(0.080)	(0.083)	(0.718)
*AI_i * $Length_i$*		-0.046***		-0.204
		(0.014)		(0.128)
*AI_i * $Quality_i$*		0.004		0.114**
		(0.004)		(0.037)
question controls	√	√	√	√
questioner controls	√	√	√	√
question timing FE	√	√	√	√

N	3,613	3,612	2,727	2,726

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. Column 2 has fewer observations than Column 1 because computing answer quality is not possible for a small number of questions affected by connection issues. Columns 3 and 4 are estimated on questions with a first human answer, and the quality outcome requires generating a reference answer, both of which reduce the sample.

The label component (H1). The AI indicator is positive and significant (0.188, $p < 0.05$). Projected to an AI answer whose content contributes nothing, the presence of a labeled AI answer raises rather than lowers the number of human answers a question attracts (i.e., AI label effect), consistent with contributors treating the AI as a weak competitor whose appearance invites a response. H1 is supported.

The length slope (H2). The interaction between the AI indicator and AI answer length is negative and significant (−0.046, $p < 0.001$). As the AI answer gets longer, the consequence of having it under the question becomes more adverse to human contribution. The magnitude is substantial relative to the pooled estimate in Column 1: a one-standard-deviation increase in $Length_i$ shifts the effect by roughly 2.502%, which is more than half the size of the average effect itself. H2 is supported.

Commented [DC3]: if I use raw length rather then log, length become not signifant at all.

The quality slope (H3). The interaction with AI answer quality is small and statistically indistinguishable from zero (0.004, s.e. 0.004). The consequence of having an AI answer under a question does not depend on how good that answer is. This is the prediction that separates our account from a capability-deterrence account, under which better AI should deter more. We did not find evidence to reject H3.

Commented [DC4]: I first do the standardization to length and quality. and there are two ways of filling value to control group sample:
1) fill NA with 0. but this means control group have average level of AI length and quality. Werid.
2) fill NA with minimum value of lengths/quality. a little meaningful but still werid.

For both ways:
1) main model do not have significant deterrence effect for length.
2) equalivilent test for length and quality fails.

Even for current log length and quality results, equilvilent test fails (p=0.09)

Together the three coefficients describe a function rather than an effect. Where the AI answer's content is thin, having an AI answer under the question draws human contribution; as content thickens, the draw weakens and then reverses.

6.2. How the Effect on Answer Quality Varies with the AI Answer

Columns 3 and 4 study the effect of internal GenAI-generated answers on quality of the first human answer.

The label component (H4). The AI indicator is non-significant (0.411, s.e. 0.718). The sign is different from the one H4 predicts: having judged the rival weak, the answerer supplies only the effort needed to beat it. However, due to the insignificance, we report H4 as not supported rather than a conclusive reject.

The length slope (H5). The interaction with AI answer length is small and insignificant (−0.204, s.e. 0.128). How much of the question the AI answer covers does not change how good the human answer is, which is consistent with our theoretical development. We did not find evidence to reject H5.

The quality slope (H6). The interaction between the AI indicator and AI answer quality is positive and significant (0.114, $p < 0.01$). The better the AI answer, the higher the quality of the human answer that follows it. This is consistent with our theoretical explanation: when AI sets a higher benchmark in the competition, human contributors respond with higher quality answers. Over the observed range of AI answer quality, roughly [5.359, 8.845] (mean ± 1 SD), this slope implies the effect of AI with a typical AI quality level on human quality is approximately [0.611, 1.008] in human answer quality score, or [0.295, 0.486] standard deviations of the outcome. H6 is supported.

6.3. The Pattern as a Whole

Table 6 sets the six tests against their predictions.

Table 6. Predictions and results

	Answer supply (entry)	Answer quality (effort)
Label component	H1: positive. 0.188, $p < 0.05$. Supported	H4: negative. n.s. No evidence of an effect
Length	H2: negative. −0.046, $p < 0.001$. Supported	H5: no effect predicted. n.s. No evidence of an effect

Quality	H3: no effect predicted. n.s. No evidence of an effect	H6: positive. 0.114, $p < 0.01$. Supported

The estimates trace the predicted pattern of Section 3.6. Length reduces supply (H2: −0.046, $p < 0.001$), and quality raises effort (H6: 0.114, $p < 0.01$). The two predictions of no effect behave as our account requires: we do not find evidence that quality shifts supply (H3, n.s.) or that length shifts effort (H5, n.s.). We regard these as an absence of evidence for an effect, not as proof that the effect is exactly zero. The label component raises answer supply (H1: 0.188, $p < 0.05$) and has no impact on human answer quality (H4, n.s.). Read together, the cue a member registers at a glance, the label and the length, governs entry, and the cue that requires reading, the quality, governs effort.

One result does most of the discriminating work: AI answer quality does not move supply (H3). This single null rules out two accounts at once. One predicts that better AI lowers human supply: under a capability-deterrence account, a stronger AI answer is a stronger rival for acceptance, so members who expect to be outperformed stay out. The other runs the opposite way: under a scaffold account, a stronger AI answer gives members a higher-quality starting point to build on and signals that the question is worth engaging, so better AI raises supply. Both let AI answer quality govern entry; they differ only in direction. The null on H3 is inconsistent with a quality slope of either sign, so it rules out the negative account and the positive one together. What survives is our timing argument: quality can be judged only by reading, and reading follows the decision to enter, so quality shapes how much effort a member invests once inside (H6) but not whether the member enters at all (H3).

The pattern also tells a platform where to intervene, because the two levers are separable. To lift participation on thinly covered questions, a platform should act on what members see before reading, for instance by adjusting whether and how prominently the AI label is shown, or by constraining or shortening the AI answer at the point of entry, since these are the cues that drive entry. To lift answer quality, a platform should instead improve the AI answer itself, since a stronger AI answer raises the effort members put in (H6) without discouraging them from answering (H3). Improving quality need not come at the cost of participation.

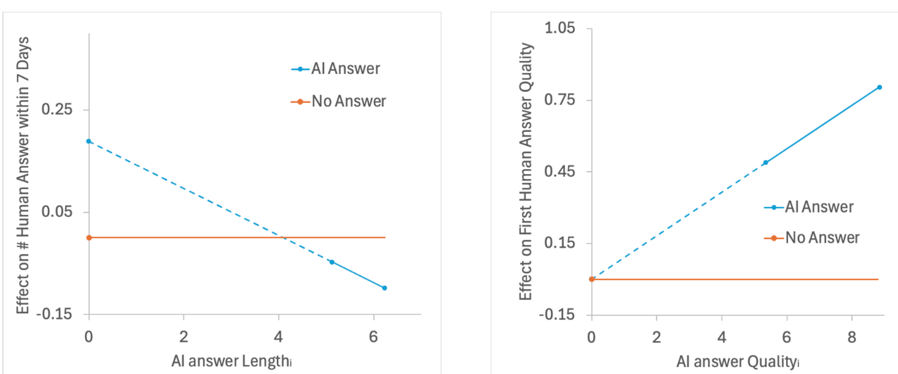
Figure 4 traces the two functions. Panel (a) plots the effect on answer supply as AI answer length varies, and Panel (b) plots the effect on answer quality as AI answer quality varies. Each panel varies only the content dimension that moves the outcome in question, because the other dimension does not: AI answer quality does not shift supply (H3), and AI answer length does not shift human answer quality (H5). In each panel the horizontal line at zero is the no-AI-answer baseline, the solid segment covers the range within one standard deviation of the mean of the varying dimension, and the dashed segment extends toward zero content to show the projected label component. Panel (a) shows a line that begins above the baseline and crosses it within the observed range; Panel (b) shows a line that begins at the baseline at zero content and rises across the observed range.

Commented [DC5]: I remove the "language about "holding the other dimension at its mean"

Because if follow this plotting, panel b would become:
1) For AI group, when AI quality is low, human quality will be lower than control group.
2) What's more, at this situation, human quality would be negative.

This may create the confusion and even speculation about out LLM measure, whose range is 0-10.

So, I think just keep its core role - illusration is fine. Focus on the story.



(a) Answer supply:
label motivates, length deters

(b) Answer quality:
higher AI quality raises human quality

Figure 4. The effect of AI-generated answers depends on both the AI label and the AI content.

Finally, a note on the reach of these results. The two slopes are the portable part of what we find, because they describe how contributors respond to features of an AI answer rather than to the AI system we happened to observe. A platform contemplating a different or a better internal GenAI can substitute that system's profile into these gradients, to infer how AI would shape human contribution under the same competition mechanism. The label component is less portable, for a reason internal to our own theory: it rests on an appraisal grounded in beliefs about AI capability, and Section 3.4.1 documents that these beliefs were, in our window, specific to a moment before practitioners had revised their assessment of large language models. As those beliefs change, the label component should be expected to change with them. We develop this in Section 7.

6.4. Robustness Checks

6.4.1. Alternative Specifications

Our main analysis uses log-transformed count data for the number of answers per question. Recent research has raised concerns about log-transforming non-negative dependent variables (Chen & Roth, 2024). We therefore re-estimate the two main relationships under models that do not require this transformation.

Answer Supply. Table 7 examines the human answer supply across five specifications. Column 1 reproduces our preferred OLS specification for comparison. Column 2 adds a questioner random effect. Because treatment is randomized at the question level, the error term is uncorrelated with treatment, so the random effect is an appropriate adjustment for repeated questions from the same questioner. Columns 3-5 employ models specifically designed for count data: Poisson, zero-inflated Poisson (ZIP), and zero-inflated negative binomial (ZINB).

The three coefficients of interest are stable across all specifications. The AI label effect is positive and significant throughout. It ranges from 0.168 ($p < 0.05$) under the random effect model to 0.469 ($p < 0.05$) under the zero-inflated negative binomial model, and the three count models sit close to 0.468. The interaction with AI answer length is negative and significant in every column, from -0.041 ($p < 0.01$) under the random effect model to -0.117 ($p < 0.01$ or better) under the count models. The interaction with AI answer quality is small and never significant, moving only between 0.003 and 0.010. Adding the questioner random effect in Column 2 leaves the OLS estimates almost unchanged. This pattern matches H1, H2, and H3.

Table 7. Robustness checks: alternative model specifications for human answer supply

	OLS	Random Effect	Poisson	ZIP	ZINB
	(1)	(2)	(3)	(4)	(5)
	DV: Log(answer number in 7 days)		DV: answer number in 7 days		
AI_i	0.188*	0.168*	0.468*	0.468*	0.469*
	(0.080)	(0.081)	(0.199)	(0.224)	(0.224)
AI_i * $Length_i$	-0.046**	-0.041**	-0.117***	-0.117**	-0.117**
	(0.014)	(0.014)	(0.034)	(0.039)	(0.039)
AI_i * $Quality_i$	0.004	0.003	0.010	0.010	0.010
	(0.004)	(0.004)	(0.014)	(0.014)	(0.014)
other controls	√	√	√	√	√
question timing FE	√	√	√	√	√
questioner random effect		√			
N	3,612	3,612	3,612	3,612	3,612
AIC		2741.982		10108.205	10110.206
BIC		3924.657			
log likelihood		-1179.991	-4864.102	-4864.102	-4864.103
deviance			3537.101		
pseudo R^2			0.014		

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. In Columns 1 and 2 the dependent variable is the log of the number of human answers provided within 7 days. In Columns 3 through 5 the dependent variable is the raw count of human answers provided within 7 days.

Human Answer Quality. Table 8 reports the OLS estimate and a specification with a questioner random effect. The interaction between the AI indicator and AI answer quality stays positive and significant in both columns (0.114, $p < 0.01$), which matches the benchmark relationship in H6. The questioner random effect does not alter the conclusion.

Table 8. Robustness checks: alternative model specifications for human answer quality

	OLS	Random Effect
	(1)	(2)
	DV: First answer quality	
AI_i	0.411	0.410
	(0.718)	(0.603)
$AI_i * Length_i$	-0.204	-0.204·
	(0.128)	(0.106)
$AI_i * Quality_i$	0.114**	0.114**
	(0.037)	(0.036)
other controls	√	√
question timing FE	√	√
questioner random effect		√
N	2,726	2,726
AIC		11959.154
BIC		13088.076
log likelihood		-5788.577

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.

6.4.2. Alternative Time Frames

Table 9 examines robustness to alternative time frames. Columns 1-4 vary the window for counting human answers from the full observation period to 3, 7, and 14 days after question posting. The results are consistent across windows. The AI label effect is positive and significant, and the interaction with answer length is negative and significant in all columns. The interaction with answer quality is not significant throughout.

Columns 5-7 restrict the sample to questions posted within the first 30, 60, and 90 days after the experiment began. Statistical power considerations are critical for interpreting these results. Our main dependent variable (number of human answers within 7 days) has a mean of 1.136 (SD: 1.093) in the treatment group and 1.305 (SD: 1.172) in the control group. Detecting the observed length slope of -0.040 with 80% power and α = 0.05 requires approximately 14,764 questions for a simple t-test. Even our regression specification includes a rich set of controls that absorb a substantial portion of residual outcome variance, which increases the precision of coefficient estimates, samples that are too small remain insufficient for reliable inference.

The 30-day window (Column 5, N=837) produces an insignificant coefficient (-0.041, $p < 0.10$), which is likely due to insufficient statistical power. The 60-day and 90-day windows (Columns 6-7, N=1,606 and 2,412) provide adequate samples and reveal significant negative effects (-0.050 and -0.044, $p < 0.01$).

Table 9. Robustness checks: alternative time frame used in analyses

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	DV: Log(answer number in the full observation period)	DV: Log(answer number in N days)			DV: Log(answer number in 7 days)		
		N=3	N=7	N=14			
AI_i	0.236*	0.167*	0.188*	0.185*	0.190	0.235*	0.212*
	(0.115)	(0.080)	(0.080)	(0.078)	(0.127)	(0.105)	(0.090)
AI_i * $Length_i$	-0.067***	-0.040**	-0.046***	-0.047***	-0.041·	-0.050**	-0.044**
	(0.020)	(0.014)	(0.014)	(0.014)	(0.023)	(0.019)	(0.016)
AI_i * $Quality_i$	0.011	0.002	0.004	0.004	0.002	0.001	-0.000
	(0.007)	(0.004)	(0.004)	(0.004)	(0.009)	(0.007)	(0.006)
question timing FE	√	√	√	√	√	√	√
other controls	√	√	√	√	√	√	√
N	3,612	3,612	3,612	3,612	837	1,606	2,412

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. In Columns 1 through 4 the dependent variable is the log of the number of human answers provided within the indicated window. In Columns 5 through 7 the dependent variable is the log of the number of human answers provided within 7 days, and the sample is restricted to questions observed for 30, 60, and 90 days.

Table 10 applies the same subsamples to the quality outcome. The interaction between the AI indicator and AI answer quality is positive in all three subsamples (0.110, 0.106, and 0.149), and its significance rises with the sample size. It is insignificant in the 30-day sample (N = 592), marginally significant in the 60-day sample (0.106, $p < 0.1$), and significant in the 90-day sample

(0.149, p < 0.01). The positive sign is stable across all three, and the increase in significance with sample size follows the same power logic that governs the supply results. The choice of observation window does not overturn the benchmark relationship in H6.

Table 10. Robustness checks: alternative time frame for human answer quality

	(1)	(2)	(3)
		DV: First answer quality	
	30 days sample	60 days sample	90 days sample
AI_i	-0.216	0.346	0.320
	(1.028)	(0.861)	(0.700)
$AI_i * Length_i$	-0.059	-0.180	-0.230·
	(0.194)	(0.155)	(0.124)
$AI_i * Quality_i$	0.110	0.106·	0.149**
	(0.074)	(0.057)	(0.045)
other controls	√	√	√
question timing FE	√	√	√
N	592	1,214	1,833

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. The sample is restricted to questions observed for at least 30, 60, and 90 days.

6.4.3. Potential Spillover Effects

On SegmentFault, AI-generated answers are visible to all users and could potentially divert attention from questions without AI answers. This contamination might influence outcomes for control questions based on the treatment status of other questions. Such spillover effects would violate the stable unit treatment value assumption (SUTVA), which requires that potential outcomes depend exclusively on a unit's own treatment status—a fundamental condition for valid causal inference in randomized experiments.

To test for potential spillover effects, we implement a comparative analysis between the five-month data of our control group and historical data from five months before the introduction of AI-generated answers (May 2023-September 2023). The two groups are both questions without AI answers and are directly comparable. Pre-experiment questions have been on the platform longer and have mechanically accumulated more answers over time. Our spillover test addresses this concern through regression controls for a cubic polynomial of question age (to substitute question/answer timing fixed effects in Equation 1, which is not applicable to the two groups on two different time slots). If spillover effects exist, control questions during the experiment period should show different outcomes compared to similar questions from the pre-AI period, as the control questions would be indirectly affected by AI answers to treated questions.

Table 11 shows no significant differences between our control group and pre-AI data for answer supply, suggesting spillover is not a concern. This likely results from the platform's interface design. Users initially see only question titles without any indication that AI-generated answers are present. AI answers are discovered only after selecting a question and viewing its detailed page. This two-step process prevents treatment status from influencing question selection, preserving the independence of potential outcomes required by SUTVA.

Table 11. Robustness to potential spillover effects

	(1)
	DV: Log(answer number in 7 days)
post-AI deployment control group indicator	-0.022
	(0.015)
controlling for a cubic polynomial of question/answer age	√
other controls	√
N	5,984

Note: 1. ***p<0.001; **p<0.01; *p<0.05.
2. We also applied all robustness checks conducted for the main effects to the spillover tests. All results remain consistent across specifications, showing no significant spillover effects.

6.4.4. Quality of Random Assignment

Since we did not conduct the experiment ourselves, the platform might have assigned AI answers based on internal metrics unobservable to us. Although our tests show no relationship between treatment status and observable characteristics, selection on unobservables remains a concern.

We employ the Oster (2019) method to assess robustness when random assignment might be compromised by unobservables. This approach examines how the treatment effect changes with observable controls and infers potential changes if unobservables were controlled. The method provides two key parameters: δ, which indicates how important unobservables would need to be relative to observables to nullify the treatment effect (values exceeding 1 suggest robustness), and β*, the bias-adjusted effect assuming selection on unobservables. These parameters require an assumption on R^2_{max}, the maximum possible R-squared value achievable if all unobservables were controlled for. R^2_{max} is typically set at 1.3 times the R^2 from the regression with observable controls only (Oster, 2019).

Table 12 reveals remarkable consistency between original and bias-adjusted coefficients. For answer supply, the original coefficient (-0.046) differs negligibly from the bias-adjusted coefficient (-0.045). The δ values for the outcomes (39.719) far exceed the recommended threshold of 1, indicating that unobservables would need to be dramatically more important than observables to explain away our findings. These results provide strong evidence for the quality of random assignment and robustness of our main conclusions.

Table 12. Robustness to the quality of random assignment

	Original coefficient β	R^2	$R^2_{\max}$	Bias-adjusted effect β^*	δ for β=0 given $R^2_{\max}$
DV: Log(answer number in 7 days)	-0.046	0.093	0.121	-0.045	39.719

Note: β and R^2 refer to our OLS coefficients and corresponding R-squared values. β^* is the bias-adjusted treatment effect. δ indicates how important unobservables would need to be, relative to observables, to explain away the estimated treatment effect.

6.5. Mechanisms

We now examine the mechanisms behind the three supported hypotheses H1, H2, and H6. The analysis proceeds hypothesis by hypothesis. For H1 we test the weak-competitor account, for H2 we test why length reduces contribution, and for H6 we test why higher AI quality raises human quality.

6.5.1. Is the AI Appraised as a Weak Competitor? (H1)

Our account for the positive label effect is that users perceive AI as a weak competitor and are therefore willing to answer. We test this account in two ways.

Test 1: question difficulty. If the weak-competitor perception is the operative mechanism, the label effect should concentrate on difficult questions. On difficult questions, a weak competitor still leaves room for a human to add value, so the label invites contribution. On easy questions, even a weak AI answer may already be sufficient, so the label should have little additional effect. We measure question difficulty using a large language model-generated score, with details in the Appendix, and we split the sample at the mean difficulty.

Table 13 confirms this prediction. On difficult questions AI label effect is positive and significant (0.282, $p<0.01$), and the interaction with answer length is negative and significant (-0.059, $p<0.001$). On easy questions AI label effect is not significant. The label motivates contribution precisely where a weak competitor still leaves a gap to fill.

Table 13. Question difficulty subgroups (H1 Test 1)

	(1) difficult questions	(2) easy questions
DV	Log(answer number in 7 days)	
AI_i	**0.282****	-0.046
	(0.099)	(0.134)
$AI_i * Length_i$	**-0.059*****	-0.017
	(0.017)	(0.024)
$AI_i * Quality_i$	0.002	0.012
	(0.005)	(0.008)
question controls	√	√
questioner controls	√	√
question timing FE	√	√
N	2,377	1,235

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. Questions are split at the mean of the difficulty score. Splitting at the median produces similar results.

Test 2: speed of the subsequent answer. A second implication of the weak-competitor account concerns timing. If a labeled AI first answer is appraised as a weak rival that invites a response, then the contribution that follows it should arrive faster when the first answer is AI than when it is human. We test this through the hazard of the answer that follows the first answer in the thread.

The test requires an event that is comparable across groups. In the treated group the first answer is the AI answer, so the contribution that follows it is the first human answer; in the control group the first answer is human, so the contribution that follows it is the second human answer. We define the event as the arrival of the contribution following the first answer in the thread, whatever its source, and we measure waiting time from the first answer rather than from the moment of posting. The clock therefore starts at the same point in both groups, and the hazard captures how quickly the next contribution arrives given that a first answer is already in place. What varies between groups is only whether that first answer was AI or human.

The waiting time is defined by a censoring rule that follows directly from how we observe each thread. For a thread in which the subsequent answer does arrive, the waiting time is the observed interval from the first answer to that subsequent answer. For a thread in which no subsequent answer has arrived, the waiting time is not observed, but the thread still carries information: we know that no subsequent answer appeared between the first answer and the point at which we crawled the data. Such a thread is right-censored, and its censoring time is the interval from the first answer to the crawl time. In both cases the quantity is measured in hours. This rule uses every thread. A thread contributes an exact waiting time when the event occurs, and it contributes a lower bound on the waiting time when the event has not yet occurred by the time of observation.

Table 14 supports the prediction. The coefficient on the AI indicator is positive and highly significant (1.012, $p < 0.001$), so an AI first answer raises the rate at which the next contribution arrives relative to a human first answer. Because the model is a Cox proportional hazards model, this coefficient is a log hazard ratio, so the implied hazard ratio is exp(1.012) = 2.75. The rate at which the next contribution arrives is therefore about 2.75 times as high when the first answer is AI as when it is human, an increase of roughly 175 percent in the instantaneous arrival rate. The effect is large. It says that an AI first answer does not merely change whether a further contribution appears but pulls it forward in time, which is what the weak-competitor account predicts.

Table 14. Effect on the speed of the subsequent answer (H1 Test 2)

	(1)
DV	Hazard of the Second Answer
AI_i	1.012 ***
	(0.077)
question controls	√
questioner controls	√
question timing FE	√
N	2,748

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. The sample includes questions with at least one answer.

6.5.2. Why Length Reduces Contribution (H2)

The length slope shows that as the AI answer gets longer, the consequence of having it under the question becomes more deterrent to human contribution. Our coverage account attributes this to breadth: a longer answer signals broader coverage and leaves less new ground for a human to cover. An alternative explanation attributes it to quality: greater length may signal a better answer that is harder to outperform. As Section 3.4.2 argues, the answerer can not assess quality at a glance and instead reads length as a cue for coverage, so we expect the length slope to reflect coverage rather than perceived quality. We test these against each other in Table 15.

Column 1 includes only the interaction with answer length, and this interaction is negative and significant (-0.044, p<0.01). Column 2 adds the interaction with answer quality. If the length effect operated through perceived quality, controlling for quality should absorb it. The length interaction is unchanged (-0.046, p<0.001), which does not support the perceived-quality explanation. Column 3 instead adds the interaction with answer coverage. We measure $AnswerCoverage_i$ as the number of bullet points in the AI answer. This count reflects the comprehensiveness and breadth of the content presented to potential contributors. The interaction with coverage is negative and significant (-0.007, p<0.05), and the length interaction falls in magnitude and drops to marginal significance (-0.028, p<0.1). Coverage absorbs much of the length effect, which supports the explanation that longer answers deter contribution because they leave less room to add something new.

Table 15. The role of answer coverage (H2)

	(1)	(2)	(3)
DV	Log(answer number in 7 days)		
AI_i	0.202**	0.188*	0.129
	(0.078)	(0.080)	(0.087)
AI_i * $Length_i$	-0.044**	-0.046***	-0.028.
	(0.014)	(0.014)	(0.016)
AI_i * $Quality_i$		0.004	
		(0.004)	
AI_i * $Coverage_i$			**-0.007***
			(0.003)
question controls	√	√	√
questioner controls	√	√	√
question timing FE	√	√	√
N	3,613	3,612	3,613

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. Column 2 has fewer observations because computing quality is not possible for a small number of questions affected by connection issues.

6.5.3. Why Higher AI Quality Raises Contribution Quality (H6)

The main results show that higher AI quality raises human contribution quality. Because AI quality is a benchmark in the competition, two processes within our account could produce this pattern. The first is improving, in which humans take the AI answer as a starting point and refine it, correcting what is weak so that the final answer is better. The second is differentiation, in which humans deliberately distinguish their answers from the AI answer. Before testing these two processes, we first rule out an alternative explanation. This alternative is selection, in which higher AI quality draws contributions from higher-quality users rather than changing how any given contributor writes. We test each in turn.

Selection. We first address the alternative explanation of selection. If selection drives the result, higher AI quality should attract more expert first human contributors. We measure the expertise of the first human contributor using the number of previously accepted answers within the relevant tag. Table 16 shows no such relationship. The interaction between AI_i and answer quality is not significant (-0.005), and the other coefficients are also insignificant. We therefore rule out selection as an explanation and turn to the two processes within our account.

Table 16. First human answer expertise and accuracy (H6, selection test and evidence from quality dimension)

	(1)	(2)
DV	Human 1 Expertise	Human 1 Accuracy
AI_i	0.048	0.852
	(0.519)	(0.726)
$AI_i * Length_i$	0.011	-0.239·
	(0.091)	(0.129)
$AI_i * Quality_i$	-0.005	0.083*
	(0.032)	(0.041)
question controls	√	√
questioner controls	√	√
question timing FE	√	√

N	2,747	2,728

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. The sample includes questions with a first human answer.

Evidence from quality dimensions. Improving and differentiation predict gains on different dimensions of the human answer, so examining where higher AI quality raises human scores lets us tell them apart. If improving drives the result, humans build on the AI answer and correct what is weak, so the gain should appear on accuracy. If differentiation drives the result, humans play to their distinctive strengths, so the gain should appear on dimensions where humans hold an edge over AI, such as personal experience. We therefore examine which dimensions of the human answer rise with AI quality. We use an LLM to score ten quality dimensions: clarity, readability, accuracy, relevance, detail level, and presence of personal experiences, insights, innovative approaches, alternative solutions, and engaging storytelling (see Online Appendix B for detailed definitions and validation). We regress each dimension of the first human answer on AI quality and report the one significant result in Column 2 of Table 16; for the other nine dimensions AI quality has no significant effect. The interaction between AI_i and answer quality is positive and significant on accuracy (0.083, $p < 0.05$): a higher-quality AI answer is followed by a more accurate human answer. This pattern is what improving predicts, that contributors examine the AI answer, locate where it is less accurate, and repair it. It also weighs against differentiation. If humans were distinguishing themselves from a strong AI answer, the gains should appear on the dimensions where humans hold a distinctive edge, such as personal experience, but AI quality has no effect on those dimensions.

Similarity between the AI and human answers. Improving and differentiation also make opposite predictions about how similar the human answer is to the AI answer. Within the treatment group, we ask whether a higher-quality AI answer is associated with greater or lower similarity to the human answer. Under improving, humans build on the AI answer, so a higher-quality AI answer should raise similarity. Under differentiation, humans move away from it, so a higher-quality AI answer should lower similarity. We measure the semantic similarity between the AI answer and the human answer using Sentence-BERT. Table 17 supports the improving explanation. The coefficient on $AIAnswerQuality_i$ is positive and significant (0.011, $p < 0.001$): as the AI answer improves in quality, the human answer becomes more similar to it, not less. This is the opposite of what differentiation predicts and confirms that the mechanism is improving. Humans do not walk away from a strong AI answer; they take it up and make it better.

Table 17. Similarity between the first and second answers (H6, improving versus differentiation)

	(1)
DV	Similarity (AI Answer, Human Answer)
$AIAnswerLength_i$	0.005
	(0.011)
$AIAnswerQuality_i$	**0.011*** **
	(0.003)
question controls	√
questioner controls	√
question timing FE	√
N	1,277

Notes. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001. Similarity is computed with Sentence-BERT. The sample includes questions with at least two answers.

7. DISCUSSION

Our study set out to understand how a labeled, platform-provided internal GenAI answer reshapes human contribution in an online Q&A community when external GenAI tools remain freely available. The central result is that the effect of internal GenAI is not one deterrence effect but the combination of a fast AI label effect and a slower AI content effect. The AI label invites human contribution because a skeptical technical community appraises the machine as a weak competitor, while the AI content pulls in the opposite direction, with length signaling coverage and suppressing supply and quality setting a benchmark that raises the quality of human answers. Because typical AI answers in our setting carry substantial content, the content effect dominates the label effect on supply and the average effect turns negative, whereas on quality the content effect operates through the benchmark and raises human quality. This decomposition is what allows our findings to be reconciled with, rather than opposed to, the one prior study of an identified AI rival contributor.

The most direct point of comparison is Su et al. (2025), who study Quora's internal GenAI and report positive effects on both the quantity and the length of human contribution, whereas we find that a longer AI answer deters human contribution and that the average effect on supply is negative. The two sets of results look contradictory on the surface, but our framework, and in particular its competition dynamics component (Chen et al. 2007; Chen 2009), shows that they are two readings of the same underlying mechanism operating under different conditions. The core of the divergence is that the AI label effect dominates in Su et al.'s (2025) setting while the deterrent effect of length is largely absent, and that the reverse holds in ours.

Competition dynamics theory holds that whether two actors are perceived as rivals depends first on their market commonality and their resource similarity, and both are low on a general-interest platform such as Quora. Quora welcomes a wide range of opinions, so a prior answer rarely forecloses further participation and market commonality between a human answerer and the AI is low. Its content is largely experiential and open-ended, where many views coexist, and the AI tends to supply objective facts while human contributors share personal experience and judgment, so the two operate on different dimensions and their resource similarity is low. Because market commonality and resource similarity are both low, the AI is not appraised as a rival but as a catalyst that revives an otherwise stalled discussion. Under this complementary framing the length of the AI answer loses its deterrent character, because a longer and more comprehensive AI answer signals more facts rather than a larger share of the space a human contributor would otherwise occupy, so greater length does not make contributors feel crowded out. The AI label reinforces this complementary reading and works in a positive direction, and the net result is that the label effect dominates while the deterrence of length does not materialize. In our technical Q&A setting the two actors instead compete for the same scarce recognition under the same question, contributions are evaluated on correctness and efficiency, precisely the dimensions on which the AI is strongest, and both market commonality and resource similarity are high. Here length is read as coverage of a finite solution space, and it deters contribution.

Timing compounds this contrast. In Su et al. (2025) the AI answer is added roughly three months after posting, once human answers have already accumulated, so the AI enters as a retrospective supplement to an established discussion. In our setting the AI answer is generated within minutes and occupies the first position before any human contributor arrives, so the human encounters the AI as a competitor at the moment of deciding whether to contribute. The same labeled AI answer therefore plays a different structural role in the two designs, and our decomposition makes clear that the sign of the aggregate effect follows from that role rather than from any fixed property of internal GenAI.

Taken as a whole, the comparison with Su et al. (2025) supports the broader claim of this paper. Whether an identified AI contributor complements or displaces human contribution is not a fixed property of internal GenAI but a joint function of the appraisal the AI invites and the signals its content carries. Reading both settings through the AI label effect and the AI content effect converts what looks like a conflict in the literature into a single conditional prediction, and it identifies the conditions, high market commonality, high resource similarity, immediate entry, and content evaluated on the dimensions where AI is strong, under which an AI rival shifts from complement to competitor.

8. CONCLUSIONS

Main results. Using a randomized field experiment at a leading technical Q&A community, we examine how a labeled, platform-provided internal GenAI answer affects human contribution when external GenAI remains available. We find that the total AI effect decomposes into a fast AI label effect and a slower AI content effect that operate in opposite directions. For answer supply, the AI label raises human contribution, consistent with a skeptical community appraising the machine as a weak competitor whose presence invites participation, while longer AI content signals coverage and suppresses contribution; because typical AI answers carry substantial content, the average effect on supply is negative. For contribution quality, the label alone does not change quality, but higher-quality AI raises the quality of the first human answer by setting a higher benchmark to which contributors respond. Our mechanism tests support this account: the label effect concentrates on difficult questions and accelerates the next contribution, coverage rather than perceived quality drives the length effect, and higher AI quality raises human quality through improving rather than through selection of stronger contributors or through differentiation from AI content.

Theoretical contribution. We contribute to the emerging literature on generative AI in online communities by studying platform-provided internal GenAI as a governance arrangement distinct from external GenAI, and by examining its consequences for human contributors in a competitive setting where external GenAI remains available. Whereas prior work studies AI that diverts questions before they arrive or that enters human answers as an unobservable input, and therefore bundles the effect of AI content with the effect of the AI label, our setting makes both the source and the content of the AI contribution visible and allows us to estimate the AI label effect separately from the AI content effect and to test the distinct decision processes through which each operates. Grounding this decomposition in dual-process theory and competition dynamics theory, we show that a community's response to an AI competitor is neither uniformly negative nor uniformly positive but depends on identifiable features of the AI answer, and we extend research on how prior contributions shape human contribution by demonstrating that the features of prior content, its presence, volume, quality, and source, do not carry the same meaning when the prior contributor is an identified machine rather than a human. The decomposition also carries a forward-looking benefit. Because our results tie the total AI effect to the content characteristics of the AI answer rather than to a fixed average, they provide a basis for reasoning about settings we do not yet observe. As AI capabilities advance and the length, quality, and other properties of AI answers shift, one can combine the AI label effect with the corresponding AI content effect to anticipate how AI will reshape human contribution under the same competition mechanism.

Practical implication. For platforms that cannot eliminate external GenAI, building an internal GenAI system is the one strategic action that is both forced upon them and fully within their power to design, and our findings speak directly to how that design should proceed. The key managerial lesson is that the label and the content of an internal GenAI answer must be managed separately, because they move human contribution in opposite directions. The visible AI label is not inherently harmful and can invite human participation, especially on difficult questions where a weak competitor still leaves room to add value, so platforms need not hide the fact that an answer is machine-generated. The deterrent force comes from content that appears to cover the question, which suggests that platforms concerned with sustaining answer volume should be cautious about displaying long, seemingly comprehensive AI answers in the first position, for example by constraining AI content in a shorter or summarized form so that it does not signal that the solution space is already occupied. At the same time, higher-quality AI raises the quality of human answers through improving rather than displacement, which implies that improving the internal GenAI need not erode human contribution quality and can instead push contributors toward the more accurate knowledge that machines do not possess. Platforms can therefore integrate capable AI while preserving the distinctive value of human contributions, provided they treat the label as an invitation and the visible extent of AI content as the lever that governs whether humans feel crowded out.

Limitations. Several limitations qualify our conclusions and point to future work. Our evidence comes from a single technical Q&A community in China, where contributions are evaluated on correctness and efficiency and where practitioners at the time held domain-specific doubts about AI capability. As our comparison with Su et al. (2025) makes clear, the sign of the aggregate effect depends on domain, on market commonality and resource similarity, and on the timing of AI entry, so the specific magnitudes we report should not be generalized to general-interest platforms or to settings where AI enters after human answers have accumulated. Our framework is intended to travel across these settings, but its predictions in domains unlike ours remain to be tested directly. The appraisal of AI as a weak competitor is also historically situated, and as beliefs about AI capability continue to revise, the AI label effect may weaken or reverse even as the content effects persist; our decomposition anticipates this shift but cannot by itself pin down when it will occur. Because external GenAI use is private and unobservable, our estimates identify the effect of adding a visible internal GenAI answer on top of a baseline in which external tools are already used, rather than the effect of AI availability in the abstract. The question-level randomization precludes any statement about question supply, and our reference-based measure of quality, though validated, captures substantive overlap with a state-of-the-art model's answer rather than every dimension of contribution value. Future research could examine how these effects evolve as AI capability and community beliefs change, extend the label-versus-content decomposition to other domains and other forms of platform-integrated AI, and study longer-run consequences for community membership and the composition of human expertise.